

0 Utility

0.1 Common Unit Conversions

- $1 \text{ atm} = 33.9 \text{ ft wg}$
- $1 \text{ gal} = \frac{1}{7.48} \text{ ft}^3$
- $1 \text{ gpm} = \frac{1}{448.8} \text{ ft}^3/\text{s}$
- $1 \text{ cfm} = 0.0004719 \text{ m}^3/\text{s}$
- $g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$
- $1 \text{ J/kg} = 1 \text{ m}^2/\text{s}^2$
- $1 \text{ psi} = 6895 \text{ Pa}$
- $1 \text{ in. wg} = 249.1 \text{ Pa at } 70^\circ\text{F}$
- $1 \text{ psi} = 27.68 \text{ in. wg}$
- $1 \text{ ft} = 12 \text{ in}, \quad 1 \text{ m} = 3.281 \text{ ft}$

2 Air Duct Systems

2.1 General Procedure

1. Assumptions:

- Maximum velocity
 - Non-industrial: maximum velocity = **1200 fpm**
 - Industrial: maximum velocity = **2200 fpm**
 - If CFM is capped, check **Table 2.3** for maximum velocity.
- If P_{fan} unknown, target pressure drop: **0.1 inch wg per 100 ft of duct**¹. Else, when pressure drop is known, calculate friction loss:

$$\text{friction loss} = \frac{P_{\text{fan}} - P_{\text{longest exit}}}{L_{\text{eq}}} \cdot 100 \text{ ft}$$
- Use the **equal friction method** to size and balance the system.
 - Aim to have equal friction loss in all branch sections of the duct system.
 - Use dampers to equalize ΔP across branches, if needed.
- **Material selection:**
 - **Sheet Metal**
 - * *Galvanized steel*: Typical for air duct system; Good corrosion resistance; cost-effective.
 - * *Aluminum*: Lightweight, corrosion-resistant; easier handling.
 - * *Stainless steel*: High corrosion resistance; suitable for high pressure and temperature.
 - **PVC (Plastic)**
 - * *ASTM D 2513*: Thermoplastic gas pressure pipe (polyethylene), for temperatures $> 428^\circ\text{F}$.
 - * *ASTM D 2517*: Reinforced epoxy resin pipe; very high corrosion resistance, pressures > 8300 psi.
- **Component Selection**
 - *Bellmouth*: Used at entrance to reduce noise.
 - *Tee 90°*: Used to split flow into two ducts.
 - *Elbow 90°*: Used to change direction of flow.
 - *Diffuser*: To distribute air. Check **Table 2.2** for pressure loss.

2. Pipe Sizing

- Use **Fig A.1** To find what you don't have. You need three of cfm, friction loss, duct size, and velocity to find the last one. Usually use cfm and friction with velocity maximums to find duct size.
- Choose **even** size if reasonable because they are more commonly available.
- Write down the diameter, cfm, friction loss in a nice table.

3. Calculate Pressure Losses

Check the main branch and then offshoots from the main branch to determine highest pressure loss.

- Calculate pressure loss due to friction:

$$\Delta P_{i-j} = (\text{friction loss}) \cdot L_{\text{eq}, i-j} \quad [\text{in. wg}]$$
 - $L_{\text{eq}, i-j}$ = equivalent length of duct section between points i and j which can be calculated by summing length of pipe and adding equivalent lengths of components from **Table A.4**.
- Calculate pressure loss due to components by referring to **Table 2.2**.
- Total pressure loss:

$$\Delta P = \Delta P_i + \sum \Delta P_{\text{components}} \quad [\text{in. wg}]$$

4. Choose a Fan:

- Select a fan that meets the required pressure drop and flow rate.
- Ensure the fan is suitable for the application (e.g., centrifugal, axial).

5. Draw final layout and balance system:

- Label each section with flow rate, duct size, length, and pressure drop.
- Add dampers and VFDs as needed for balancing flow across branches.

6. Conclusion

- Ensure all components are properly sized and selected.
- Verify that the system meets design criteria for flow rate, pressure drop, and noise levels.
- Discuss improvements

¹Most common case.

2.2 Reynolds Number Concepts

2.2.1 Internal Flow and Flow Regimes

- **Laminar flow:** Smooth, ordered motion.
- **Turbulent flow:** Disordered, fluctuating motion.
- Characterized by the Reynolds number:

$$\text{Re} = \frac{\rho V D_h}{\mu} \quad \text{or} \quad \text{Re} = \frac{V D_h}{\nu}$$

- For non-circular ducts, use hydraulic diameter:

$$D_h = \frac{4A_c}{P}$$

(where A_c is cross-sectional area, P is perimeter)

Hydraulic Diameters for Common Duct Shapes:

- Square duct ($L \times L$):

$$D_h = \frac{4(L \cdot L)}{4L} = L$$

- Rectangular duct ($L \times w$):

$$D_h = \frac{4Lw}{2L + 2w} = \frac{2Lw}{L + w}$$

- Annular duct (between outer D_2 and inner D_1):

$$D_h = \frac{D_2^2 - D_1^2}{D_2 + D_1} = D_2 - D_1$$

Note: For flow rate, use actual area not D_h .

2.2.2 Flow Characterization

- Use critical Reynolds number: $\text{Re}_{\text{cr}} = 2300$
- $\text{Re} < \text{Re}_{\text{cr}}$: Laminar flow
- $\text{Re} > \text{Re}_{\text{cr}}$: Turbulent flow

2.3 Head Losses

2.3.1 Major and Minor Losses

- Major losses (H_l): Caused by wall friction and viscous effects in fully developed flow through straight duct sections.
- Minor losses (H_{lm}): Result from components such as entrances, exits, bends, fittings, valves, and sudden area changes. In air ducts, this includes filters and heating/cooling coils.
- **Units:** h_l, h_{lm} in J/kg or m^2/s^2 (energy head); H_l, H_{lm} in m or ft (elevation head)
- Total head loss:

$$H_{lT} = H_l + H_{lm} = \frac{h_l + h_{lm}}{g}$$

2.3.2 Major Head Loss Formula

$$h_l = f \cdot \frac{L}{D} \cdot \frac{V^2}{2} \quad \text{or} \quad H_l = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

- For **laminar flow**:

$$f = \frac{64}{\text{Re}}$$

- For **turbulent flow**, use the Moody chart (default method):

1. Calculate Re
2. Determine relative roughness ε/D (from Table A-1)
3. Use Re and ε/D to find f on the chart
4. Compute head loss using f

- Alternatively, use:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\frac{6.9}{\text{Re}} + \left(\frac{\varepsilon/D}{3.7} \right)^{1.11} \right] \quad (\text{Haaland})$$

2.3.3 Minor Head Loss Formula

$$h_{lm} = K \cdot \frac{V^2}{2} \quad \text{or} \quad H_{lm} = K \cdot \frac{V^2}{2g}$$

$$h_{lm} = f \cdot \frac{L_{\text{equiv}}}{D} \cdot \frac{V^2}{2}$$

- K = loss coefficient (from tables)
- L_{equiv} = equivalent length of fitting/device

2.4 Rectangular Duct Sizing and Aspect Ratio

- Use hydraulic diameter for Reynolds number and friction calculations:

$$D_h = \frac{2Lw}{L + w}$$

- Cross-sectional area:

$$A = L \cdot w$$

- **Aspect Ratio (AR):**

$$\text{AR} = \frac{L}{w} \quad (\text{smaller is better})$$

– Recommended: $\text{AR} < 4$

– If $\text{AR} > 4$:

- * Higher frictional losses (more perimeter for same area)
- * More expensive fabrication

- **Design Steps:**

1. Use circular equivalent method to determine round duct size based on Q and acceptable V (see Fig. A.1)
2. Select a rectangular duct with same area and acceptable aspect ratio (see Table A.3)
3. Check space, noise, and architectural constraints

2.5 Variable Definitions

- Re : Reynolds number
- D_h : Hydraulic diameter
- A_c : Cross-sectional area
- P : Perimeter
- f : Darcy friction factor
- ε : Roughness
- V : Average velocity
- L : Duct length
- D : Diameter
- g : Gravitational acceleration
- h_l, h_{lm} : Head loss (J/kg)
- H_l, H_{lm} : Head loss (m)
- K : Loss coefficient
- L_{equiv} : Equivalent length of fitting

3 Liquid Pipe System

1. Define system and assumptions:

- Identify known values (e.g., Q , pipe dimensions, velocity, elevation changes).
- Set fluid properties (e.g., ρ , μ at given temperature). **Table HEX liquid-gas on Canvas.**
- List all fittings, joints, valves with K values or equivalent lengths.
 - **Flanged fittings** are popular
- **Schedule 40 steel** is popular (low carbon steel).
- Typical velocities can be found in **Table A.12.**
 - **Cold water supply piping:** 4-8 ft/s
 - **Pump suction line:** ≤ 5 ft/s
- Check erosion limits in **Table A.13.**
- Target pressure drop in piping: ≤ 3 ft wg/100 ft
- Recommended to convert gpm to ft^3/s for calculations.
 $1 \text{ gpm} = 1/448.8 \text{ ft}^3/\text{s}$

2. Sizing

- If pipe diameter is not given, check **Fig. A.3** or **Fig. A.4**, depending on material.
 - **Open system** if the system is open to the atmosphere.
 - **Closed system** if the system is closed to the atmosphere.

3. Apply the energy equation:

$$H_{\text{pump}} = \frac{w_{\text{pump}}}{g} = H_{IT} + \left[\frac{P_2}{\rho g} + \alpha_2 \frac{V_{\text{ave},2}^2}{2g} + z_2 \right] - \left[\frac{P_1}{\rho g} + \alpha_1 \frac{V_{\text{ave},1}^2}{2g} + z_1 \right]$$

$\alpha = 2$ for laminar flow, $\alpha = 1.06 \approx 1$ for turbulent flow

4. If ① and ② are:

- Both upstream or both downstream of the pump: $H_{\text{pump}} = 0$
- On opposite sides of the pump: $H_{\text{pump}} \neq 0$

5. Compute head losses:

- **Major loss:**

$$H_{\text{major}} = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

- Find f using the Moody chart with ε/D (see **Table A.1**), or use the Swamee-Jain

equation:

$$f = 0.25 \left[\log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{\text{Re}^{0.9}} \right) \right]^{-2}$$

- Other friction factor formulas are listed in **Table A.2.**

– If Using Sizing Chart:

$$H_l = \frac{(\text{loss per 100 ft})}{100} \cdot L_{\text{pipe}} \quad [\text{ft}]$$

- * Used when referencing **Table A.3** or **Table A.4** for direct friction loss values.

- **Minor loss:**

$$H_{\text{minor}} = \sum K \cdot \frac{V^2}{2g}$$

- Use K values from **Table A.14.**
- Entrance losses: $K_{\text{inlet}} = 0.78$ (from notes); sometimes $K_{\text{inlet}} = 0.5$ and $K_{\text{outlet}} = 1$ may be used.

- **Total head loss:**

$$H_{IT} = H_{\text{major}} + H_{\text{minor}}$$

6. Determine flow parameters:

- Compute V_{ave} :

$$V_{\text{ave}} = \frac{Q}{A} = \frac{4Q}{\pi D^2}$$

- Compute Reynolds number:

$$\text{Re} = \frac{4\rho Q}{\pi \mu D} = \frac{V_{\text{ave}} \rho D}{\mu}$$

- Estimate friction factor f :

- Use Moody chart if ε/D is known.
- Or apply Haaland equation:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\frac{6.9}{\text{Re}} + \left(\frac{\varepsilon/D}{3.7} \right)^{1.11} \right]$$

7. Compute pump power:

$$\text{bhp} = \frac{\rho g Q H_{\text{pump}}}{\eta_{\text{pump}}} \Rightarrow \text{Convert to hp as needed}$$

or efficiency:

$$\eta_{\text{pump}} = \frac{\rho Q g H}{\omega T_{\text{shaft}}} = \frac{W_{\text{water horsepower}}}{\text{bhp}} = \frac{\rho Q g H}{\text{bhp}} = \frac{mgH}{\text{bhp}}$$

8. Cavitation Check (NPSH):

- Use energy equation to find pressure at pump inlet:

$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_{\text{ave},1}^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_{\text{ave},2}^2}{2g} + z_2 + H_{IT, \text{suction}}$$

Find $H_{IT, \text{suction}}$ using the same method as above.

- Compute:

$$\text{NPSHA} = \left(\frac{P}{\rho g} + \frac{V_{\text{avg}}^2}{2g} \right)_{\text{pump inlet}} - \frac{P_{\text{vapor}}}{\rho g}$$

vapor pressure can be found in **Table A-9E**.

- Compare with NPSHR to ensure:

$$\text{NPSHA} > \text{NPSHR}$$

4 Heat Exchangers

4.1 General Procedure for Finned-Tube Heat Exchanger Design

1. Define design goals and assumptions:

- Specify target heat transfer rate (e.g., 96,000 Btu/hr), air inlet/outlet temperatures, and water conditions.
- Choose a reasonable air velocity over the coil.
Common practice:
 - $V_{\text{air}} = 400\text{--}500$ fpm to reduce noise and align with filter performance.
 - 500 fpm is suitable for residential heating coils (see Table 2-1).
- Choose a water velocity in tubes:
 - $V_{\text{water}} \approx 4$ fps is typical for potable or building service water.
 - Below copper erosion limits and compatible with pump suction velocity constraints.
- Use typical material choices:
 - Tube material: **Type K copper** (high conductivity, standard availability).
 - Fin material: **Copper** for improved heat transfer.
- Select standard tube geometry:
 - Outer diameter: $D_o = 0.402$ in (approx. 3/8 in. nominal).
 - Inner diameter: $D_i = 0.305$ in (based on Type K 1/4 in NPS, Table A.6).
 - Tube pitch: $x_L = 0.866$ in (longitudinal), $x_t = 1.0$ in (transverse) [Figure C.4a].
- Use a **counterflow** configuration:
 - Higher effectiveness compared to parallel flow.
 - Staggered tube arrangement increases turbulence and enhances heat transfer.

2. Use ϵ -NTU method to find surface area:

- Determine NTU from effectiveness and capacity ratio (charts or equations).

$$NTU = \frac{UA_o}{C_{\min}} \Rightarrow A_o = \frac{NTU \cdot C_{\min}}{U}$$

3. Determine heat transfer coefficients:

- Calculate $\bar{h}_i$ using Dittus-Boelter (for water in tubes):

$$\text{Re}_D, \text{Pr}, \quad Nu = 0.023 \cdot \text{Re}^{0.8} \cdot \text{Pr}^{0.3}, \quad h = \frac{Nu \cdot k}{D}$$

- Calculate $\bar{h}_o$ from j -factor charts (air side).
- Determine fin efficiency η_o and surface ratio A_i/A_o .
- Combine into overall U :

$$\frac{1}{U} = \frac{1}{\eta_o \bar{h}_o} + R_{fo} + \frac{A_i}{A_o} \left(R_{fi} + \frac{1}{h_i} \right)$$

4. Find coil dimensions:

- Use:

$$\Omega = \frac{A_o}{\alpha}, \quad W = \frac{\Omega}{A_f}$$

- Number of tube rows:

$$N_r = \frac{W}{x_L}, \quad \text{round up}$$

- Number of tubes per row:

$$N = \frac{C_{\min}}{\rho_w c_{p,w} V_w \cdot (\pi D_i^2 / 4)}$$

- Dimensions:

$$H = N \cdot x_t, \quad L = \frac{A_f}{H}$$

5. Compute air and water pressure losses:

- Use j - and f -factor charts and formulas from textbook.
- Convert to in. wg using air or water density at mean temperature.